

BEYOND MEAN FOILS: AUDITING WORST-FOIL SPECIFICITY IN FROZEN CLIP REGION EXPLANATIONS

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ABSTRACT

A frozen CLIP region can pass locality and mean-foil screens while responding more strongly to a non-target foil. We audit the maximum-target-drop single CCI region using matched prompt-mean and aggregate worst-foil margins. In the frozen COCO B/16 archive, 41.18% of eligible images pass both screens yet fail the worst-foil audit; 64.38% [63.67, 65.07]% of screen-passers fail. Target-tolerant reranking at $\epsilon = .02$ switches 2.0–2.6% of images; paired margin gains persist under category-disjoint and annotation-absent foils. Independent four-setting MPS reruns add direct paired gains over Mean and Max-.1, but gains over Max-.1 are small. Switched-image bbox precision drops by 4.95 and 13.20 percentage points on COCO and VOC B/16. Margin improvement is sparse, differs from sign repair, and does not imply loss-free explanations.

Index Terms— CLIP, explainability, interventions, model auditing, worst-foil specificity

1 INTRODUCTION

Frozen vision-language models such as CLIP [1] are often interpreted through intervention-induced image–text score changes. CCI [2] clusters patch tokens and combines cluster importance into a heatmap. We audit only its maximum-target-drop region (CCI-top1). A large target drop need not be target-specific: generic object evidence can support another class more strongly, despite high box overlap. Averaging easy foils can hide this failure. In the frozen COCO B/16 archive, 11,411 of 17,725 locality/mean-class screen-passers (64.38%) have negative worst-foil margin.

Our contribution is the selected-region response-vector audit and controlled evidence of local improvement, not the max/CVaR operator [3]. COCOA [4] defines corpus-versus-foil attribution; CASE [5] diagnoses class-insensitive saliency and builds contrastive explanations. We test what locality and mean-class screens miss, then separate region selection from held-out audit. Related CLIP attribution [6, 7, 8], class/concept contrasts [9, 10, 11] and caption foils [12] motivate this distinction. Image Description Enhanced CLIP-Adapter (IDEA) [13]

combines visual features and descriptions for few-shot classification; our audit keeps the model fixed. Reranking probes local improvement. Negative relative specificity does not imply absent target evidence: one region may support several classes.

2 WORST-FOIL AUDIT AND TARGET-TOLERANT RERANKING

For checkpoint m , $s_m(x, c) = \tau_m \langle z_m(x), e_m(c) \rangle$ uses unit-normalized embeddings and the checkpoint’s learned $\tau_m = \exp(\text{logit_scale}_m)$, before softmax. For candidate region R and concept prompt c , define

$$d_R(c) = s_m(x, c) - s_m(\mathcal{I}(x, R), c), \quad (1)$$

where $\mathcal{I}$ is a frozen internal intervention. Our single-region CCI baseline selects

$$R_{\text{CCI}} = \arg \max_{R \in \mathcal{R}(x)} d_R^{\text{sel}}(t), \quad (2)$$

using only the selection prompt. CCI-top1 denotes this baseline; the CCI subscripts and figure labels retain this shorthand. For non-target foils $\mathcal{F}_t$, a worst-foil margin compares the target response with the strongest foil response.

Matched held-out estimands. Selection uses “a photo of a {category}”. Held-out evaluation uses “a picture of the {category}”, “an image containing a {category}”, and “the {category}”. Let $e_h(c)$ denote the stored unit-norm text embedding for held-out prompt h , and $n_c = \|H^{-1} \sum_h e_h(c)\|_2$. We apply the same class-specific normalization rule, $\tilde{d}_h(c) = d_h(c)/n_c$ and define

$$\begin{aligned} P_{\text{max}}^{\text{pm}}(R, t) &= H^{-1} \sum_h [\tilde{d}_h(t) - \max_f \tilde{d}_h(f)], \\ P_{\text{max}}^{\text{agg}}(R, t) &= H^{-1} \sum_h \tilde{d}_h(t) - \max_f H^{-1} \sum_h \tilde{d}_h(f). \end{aligned} \quad (3)$$

Thus $P_{\text{max}}^{\text{pm}} \leq P_{\text{max}}^{\text{agg}}$: the former permits a different hardest foil per template. Both are ontology-relative specificity measures. For the mean-class screen, write $\bar{d}_c = H^{-1} \sum_h \tilde{d}_h(c)$.

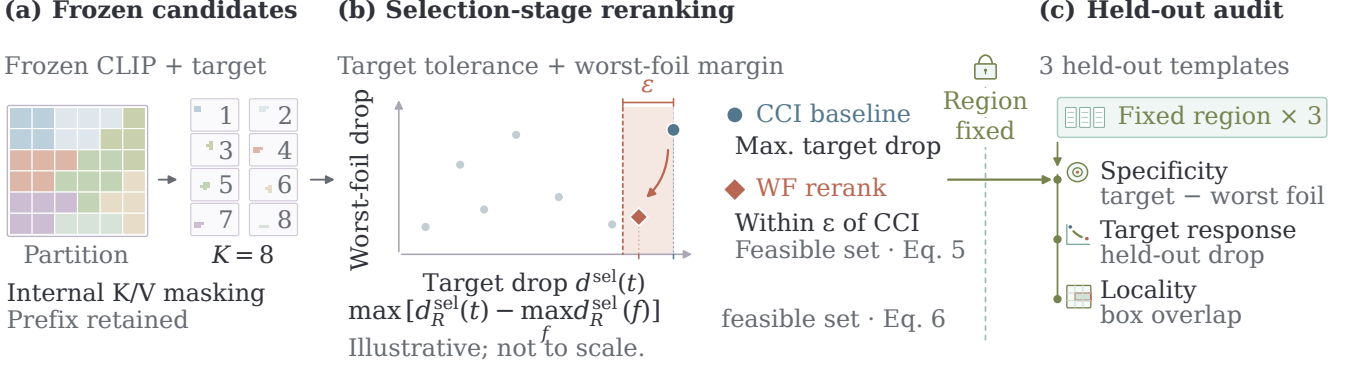


Figure 1: Selection and held-out audit. Frozen CCI candidates are selected or reranked only from selection-stage tensors. The region is fixed before held-out specificity, target-response, and locality evaluation. The candidate-response diagram is illustrative and not to scale.

The archived screen uses $P_{\text{mean,all}} = \bar{d}_t - |\mathcal{C}|^{-1} \sum_{c \in \mathcal{C}} \bar{d}_c$ over all dataset classes, equally weighted, on these same held-out responses. Excluding the target gives $P_{\text{mean,foil}}$, with $P_{\text{mean,all}} = (|\mathcal{C}| - 1)P_{\text{mean,foil}}/|\mathcal{C}|$; their positive screens coincide. This class average differs from prompt-mean.

Selection-stage target tolerance. For each candidate define

$$M^{\text{sel}}(R, t) = d_R^{\text{sel}}(t) - \max_{f \in \mathcal{F}_t} d_R^{\text{sel}}(f). \quad (4)$$

Given tolerance ϵ , form

$$\mathcal{R}_\epsilon = \{R : d_R^{\text{sel}}(t) \geq d_{R_{\text{CCI}}}^{\text{sel}}(t) - \epsilon\}, \quad (5)$$

and rerank only this feasible set:

$$R_{\text{WF}}(\epsilon) = \arg \max_{R \in \mathcal{R}_\epsilon} M^{\text{sel}}(R, t). \quad (6)$$

Because $R_{\text{CCI}} \in \mathcal{R}_\epsilon$, the feasible set is nonempty; any switch loses at most ϵ of the *selection-stage* target drop. This is not a held-out target guarantee. Equation 6 reads only selection-stage tensors, and held-out prompts are used only after the region is fixed. We use $\epsilon = .02$ as a fixed operating point, not an optimized value. Its cosine-scale tolerance is $.02/\tau_m$, not necessarily equal across checkpoints. Figure 1 summarizes the separation.

3 EXPERIMENTAL PROTOCOL

Models, intervention, and endpoints. We use $K = 8$ frozen CCI regions on COCO val2014 [14] (27,708 eligible images) and VOC2007 test [15] (1,943), with OpenAI CLIP B/16 and B/32. Reranking changes only which frozen candidate is selected. Each image has one target: the largest-area annotated instance’s category (COCO annotation area; VOC box area). Eligibility requires another annotated category and target-instance area between 2% and 80% of the image; COCO crowd annotations are excluded. Full foils comprise every other dataset category (79 for COCO, 19 for VOC). The intervention blocks selected patch key/value columns in every

visual-transformer block and head by adding $-\infty$ before attention softmax, while retaining the prefix token. Cached patch features are clustered by K-means ($n_{\text{init}} = 3$, 50 iterations, seed $1701 + i$ for cache index i). BBox precision is the fraction of selected patch centers in the union of transformed boxes of the target category, not only its largest instance, and is reported separately because perturbation metrics depend on the intervention operator [16]. Ties select the first exact argmax.

Held-out controls and uncertainty. Selection uses one template and evaluation uses the three held-out templates in Eq. 3. Cross-foil controls use ten fixed seeds 1701, 2701, ..., 10701 to permute categories into equal alternating folds. Each direction selects using one fold and audits on the other, excluding the target from both. The 20 seed–direction deltas are averaged within each image before paired unique-image resampling; annotation-absent evaluation removes categories annotated as present without claiming semantic absence. Fixed 5/19-foil subsets probe cardinality and composition jointly. Intervals use 10,000 paired unique-image bootstrap resamples; the archive fixes the base seed at 1701 and records deterministic setting/metric-specific seeds. Only four provenance-valid OpenAI settings enter the direct reranking analysis; two historical LAION settings are excluded because their text-normalization provenance is unresolved.

Independent local rerun. We regenerated candidates and responses on a MacBook M4 Pro using PyTorch 2.8.0 MPS, the same declared OpenAI models, splits, templates, $K = 8$ and $\epsilon = .02$. Tables 2, 3 and 6 report this rerun; Table 1, foil/tolerance controls and Figures 2–3 retain frozen-archive results. Rerun intervals are 10,000-draw percentile image bootstraps with recorded metric-specific seeds. Direct comparisons resample same-image differences; conditional failure jointly resamples numerator and denominator, and switch intervals resample switched images. Intervals are pointwise, without multiple-comparison adjustment.

Table 1: Frozen archive: CCI-top1 vs. WF at $\epsilon = .02$. $P = P_{\max}^{\text{agg}}$; brackets are paired 95% CIs; “Neg.” is baseline→WF negative-margin rate.

Setting	P_{CCI}	ΔP	Δd_t	Δbbox	Sw.	Neg.
COCO B/16	-.342	+0.0150 [+.0131,+.0169]	+0.00005 [-.00034,+.00044]	-.00118 [-.00199,-.00036]	2.39%	74.04→73.89%
COCO B/32	-.304	+0.0108 [+.00945,+.0122]	-.00010 [-.00046,+.00027]	-.00080 [-.00162,+.00002]	2.21%	73.91→73.75%
VOC B/16	+.168	+0.0118 [+.00677,+.0183]	-.00085 [-.00227,+.00059]	-.00340 [-.00652,-.00034]	2.57%	58.16→57.59%
VOC B/32	+.186	+0.00867 [+.00441,+.0143]	-.00069 [-.00184,+.00040]	+0.00023 [-.00288,+.00341]	1.96%	55.89→55.58%

Table 2: Local rerun CCI-top1 screen audit. A: bbox $\geq .5$ and mean-foil margin > 0 ; F: worst-foil margin < 0 . Rates and conditional 95% CIs in %.

Setting	$\Pr(A)$	$\Pr(A \cap F)$	$\Pr(F A)$
COCO B/16	63.97	41.18	64.37 [63.67,65.05]
COCO B/32	64.41	41.72	64.78 [64.07,65.47]
VOC B/16	62.22	26.30	42.27 [39.45,45.05]
VOC B/32	63.15	25.99	41.16 [38.40,43.91]

4 RESULTS

Worst-foil debt. Frozen COCO B/16 CCI-top1 has $P_{\max}^{\text{agg}} = -.342$ [-.357,-.327], raw target response .939 [.924,.953] and bbox precision .679 [.674,.683]. These are full-sample means. Within its negative-margin subset, mean target drop is .481 and 90.40% have positive drop. The independent four-setting audit (Table 2) finds COCO B/16 failures in 11,410/17,726 screen-passers, versus frozen 11,411/17,725.

Sparse margin improvement. Frozen Table 1 shows positive aggregate margin gains in all settings. Prompt-mean gains are +.0150, +.0108, +.0119, +.00867, also with CIs above zero. All raw target-response CIs include zero, without establishing equivalence. B/16 bbox declines on COCO and VOC; B/32 intervals include zero.

The gain is not merely selector–endpoint reuse. Frozen cross-foil controls (Figure 2) retain positive normalized aggregate gains: .01052/.00728/.00705/.00438 across COCO B/16, COCO B/32, VOC B/16 and VOC B/32. Annotation-absent evaluation gives .00939/.00697/.00809/.00443. All paired 95% CIs exceed zero. These controls establish gain persistence, not restricted-foil baseline conditional failure rates.

Direct same-feasible-set comparisons. Table 3 reports direct rerun WF-minus-alternative differences. Mean maximizes $d_t^{\text{sel}} - \text{mean}_f d_f^{\text{sel}}$ and Max-.1 maximizes $d_t^{\text{sel}} - .1 \max_f d_f^{\text{sel}}$ over the same $\mathcal{R}_\epsilon$. All eight pointwise margin CIs exceed zero, but gains over Max-.1 are small. Relative to Mean, bbox declines on COCO B/16, COCO B/32 and VOC B/16 with CIs

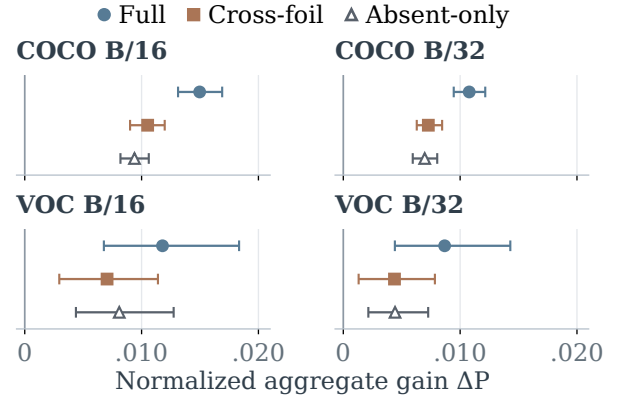


Figure 2: Held-out normalized aggregate gains at $\epsilon = .02$ for full, category-disjoint cross-foil, and annotation-absent evaluation; bars are paired 95% CIs.

below zero. Relative to Max-.1, bbox CIs include zero in all settings, while COCO B/16 raw target response decreases by .000099 [-.000192, -.000010]. These results do not establish loss-free superiority.

Why switching is sparse. In the local rerun, singleton feasible sets account for 95.29/95.60/95.21/95.27% of COCO B/16, COCO B/32, VOC B/16 and VOC B/32 images. The counts with 2/3/4 candidates are 1215/81/8, 1174/43/2, 88/5/0 and 89/3/0; none have more than four. COCO B/16 permits an alternative in 1,304/27,708 images, and 663/1,304 (50.84%) switch. The frozen counterpart is 1,303.

The effect also reduces the breadth of foil violations. As a secondary held-out diagnostic, let $\bar{d}(c) = H^{-1} \sum_h \bar{d}_h(c)$ for the already fixed region. We report V_{rate} , the fraction of foils with $\bar{d}(f) > \bar{d}(t)$, and $V_{\text{mass}} = |\mathcal{F}_t|^{-1} \sum_{f \in \mathcal{F}_t} [\bar{d}(f) - \bar{d}(t)]_+$, averaging over all foils, including zeros. On COCO/OpenAI B/16, reranking at $\epsilon = .02$ changes V_{rate} from .18973 to .18751 and V_{mass} from .05878 to .05708. Violation breadth therefore also decreases descriptively.

Foil-cardinality sensitivity. Adding foils can increase the maximum response, but the fixed subsets in Table 4 vary cardinality and composition jointly. The reranking gain stays

Table 3: Local rerun direct paired $\Delta P_{\max}^{\text{agg}}$, multiplied by 10^3 ; pointwise 95% CIs. Each column subtracts the named alternative from WF.

Setting	WF-Mean	WF-Max-.1
COCO B/16	12.461 [10.838,14.193]	0.245 [0.131,0.372]
COCO B/32	9.343 [7.970,10.807]	0.241 [0.088,0.418]
VOC B/16	4.480 [2.332,7.019]	0.507 [0.025,1.114]
VOC B/32	3.184 [1.450,5.126]	0.418 [0.044,0.907]

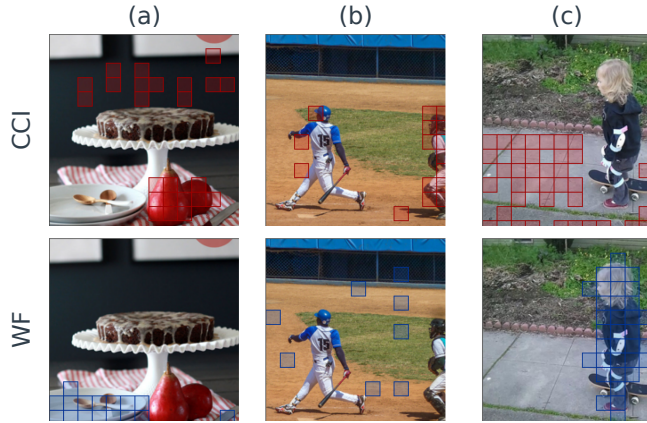


Figure 3: Archived rule-selected COCO/OpenAI B/16 cases at $\epsilon = .02$ (top/bottom: CCI/WF); ΔP is normalized aggregate gain: (a) locality gain, dining table, $\Delta P = +4.58$; (b) specificity-locality trade-off, person, $\Delta P = +0.78$; (c) counterexample, person, $\Delta P = -0.98$. Bbox precision: $.46 \rightarrow 1.00$, $.81 \rightarrow .11$, and $.02 \rightarrow .88$, respectively.

Table 4: Frozen archive: COCO foil-count sensitivity. “Neg.” is baseline negative-margin rate; ΔP is normalized aggregate gain.

Setting	Neg.@5	Neg.@19	Neg.@all	ΔP_5	ΔP_{19}
COCO B/16	45.8%	63.5%	74.0%	+0.00872	+0.01074
COCO B/32	45.0%	62.8%	73.9%	+0.00841	+0.00947

positive at both matched counts, so the effect is not created solely by evaluating against a larger foil set.

Cross-template consistency. Frozen mean pairwise hardest-foil agreement is 59.80%. All three templates agree on category in 46.36% of images and on the stored semantic family in 53.69%; the latter two use the all-three criterion.

Tolerance sensitivity. The frozen frontier (Table 5) increases switches with tolerance; $\epsilon = .02$ was fixed independently. At $.20$, full-sample target/bbox deltas are $(-.01396, -.01627)$, exposing the target-specificity trade-off.

Switch-conditioned outcomes. Table 6 uses rerun switched-image vectors directly. VOC B/16 loses 13.20 bbox percentage points $[-24.62, -1.80]$. COCO B/16 loses 4.95

Table 5: Frozen archive: COCO/B16 tolerance sensitivity; ΔP is full-sample gain, while $\Delta P \mid \text{sw}$ and $\text{Pr}(+)$ condition on switches.

ϵ	Sw.	ΔP	$\Delta P \mid \text{sw}$	$\text{Pr}(+)$
.01	1.23%	+0.00791	+0.642	90.9%
.02	2.39%	+0.01497	+0.626	92.2%
.05	6.24%	+0.03906	+0.626	91.6%
.10	11.68%	+0.07643	+0.654	92.0%
.20	21.87%	+0.14479	+0.662	92.9%

Table 6: Local rerun switch-conditioned WF-CCI outcomes at $\epsilon = .02$. Raw target and bbox means with 95% switched-image CIs; bbox multiplied by 100 (percentage points).

Setting	n_{sw}	Δd_t	$100\Delta \text{bbox}$
COCO B/16	663	0.002 [-0.013,0.018]	-4.95 [-8.33,-1.45]
COCO B/32	614	-0.004 [-0.021,0.012]	-3.61 [-7.45,0.05]
VOC B/16	50	-0.032 [-0.086,0.020]	-13.20 [-24.62,-1.80]
VOC B/32	38	-0.035 [-0.091,0.021]	1.16 [-15.00,17.26]

points $[-8.33, -1.45]$, with median zero and quartiles $-.3242, +.1137$. Raw target-response intervals include zero in all four switch subsets; this is not an equivalence result. Margin improvement occurs in 611/663, 532/614, 46/50 and 35/38 switches, whereas sign repair occurs in only 42/663, 48/614, 11/50 and 6/38.

Cases and uncertainty. Figure 3 retains archived selected cases, not frequencies. All 59,302 rerun image-model records pass strategy/array checks. CPU/MPS smoke checks cover only 1–2 images per setting.

Interpretation and scope. Rerun and archive are separate evidence sets. COCO B/16 image IDs align, but three images change screen/failure status and ten change CCI region labels. Some response changes exceed threshold rounding; frozen candidate tensors are unavailable for causal attribution. COCO B/32 switches 614 times versus 613 in the archive, whose missing per-image fields prevent localization. These differences are not attributed to MPS or treated as exact reproduction. Inference is conditional on eligible images, templates, foils, candidates and intervention. Restricted-foil conditional failure rates and robustness to new candidate or intervention configurations remain untested.

Conclusion. CCI-top1 auditing exposes specificity failures and sparse margin improvement. Independent reruns support positive paired margin gains over Mean and Max-.1 under the tested protocol, with small gains over Max-.1 and measurable locality costs. Target response, locality and sign repair remain distinct endpoints.

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